

Interpretable Multimodal Brain Age Prediction via Cauchy-Schwarz Divergence-Induced Gács-Körner Common Information

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Abstract

Multimodal brain age prediction requires the principled fusion of heterogeneous neuroimaging modalities while preserving interpretability and robustness. However, existing fusion strategies often entangle redundant and modality-specific information, rely on restrictive distributional assumptions, or lack a clear theoretical foundation. In this paper, we propose CS-GKDR, an information-theoretic framework that explicitly decomposes multimodal representations into shared and modality-specific components for brain age prediction. Our method generalizes the Gacs-Korner common information to the multivariate and continuous setting, and introduces Cauchy-Schwarz divergence induced information measures to obtain a tractable and scalable learning objective. This formulation avoids variational approximations, Gaussian assumptions, and discretization, making it well-suited for high-dimensional and structured neuroimaging data. Experiments on two large-scale multimodal neuroimaging datasets demonstrate that CS-GKDR consistently outperforms state-of-the-art multimodal baselines in predictive accuracy. Moreover, the extracted information components provide biologically interpretable insights into the relative roles of structural, functional, and demographic factors in brain aging.

Overall Framework

The overall construction pipeline of CS-GKDR is summarized in Figure 1.

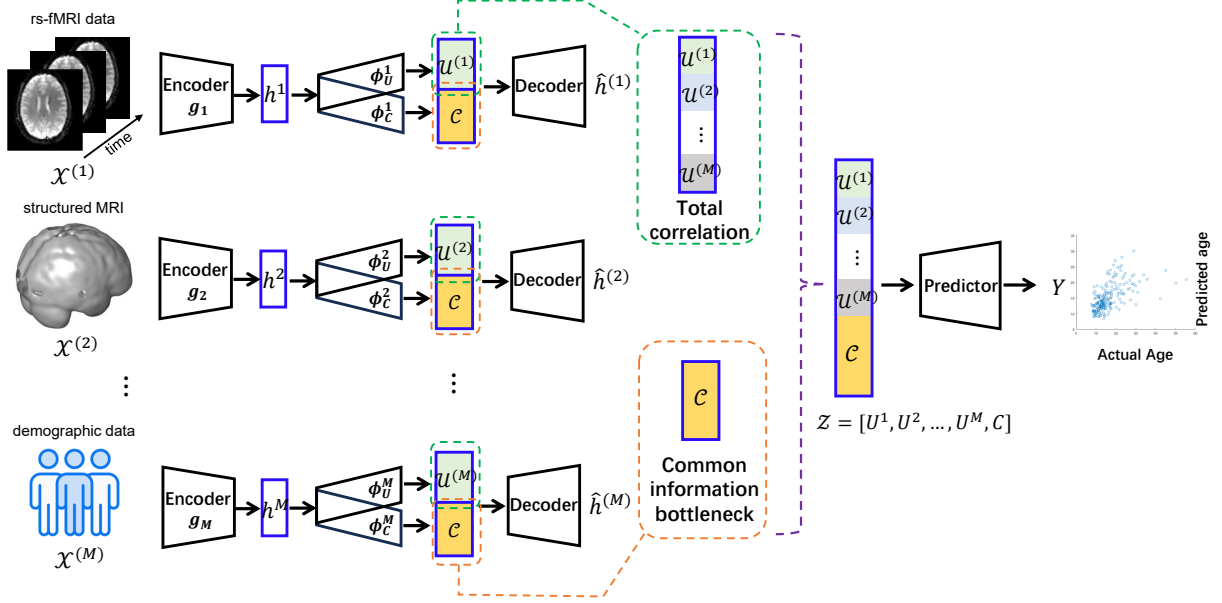


Figure 1: Overall Framework of CS-GKDR.

We first encode each modality X_m (with samples x_m , $m = 1, \dots, M$, where M denotes the number of modalities) into features: $h_m = g_m(x_m)$, where $g_m(\cdot)$ is the encoder for the m -th modality, and h_m is the latent representation. We then explicitly decompose the representation into a shared representation C (cross-modal common information) and a modality-specific representation U_m (unique information coming only from the m -th modality), and reconstruct at the feature level via a decoder: $\hat{h}_m = d_m(U_m, C)$, where $d_m(\cdot)$ is the decoder for the m -th modality, and $\hat{h}_m$ is the reconstructed feature, thereby ensuring that (U_m, C) preserves the effective information of that modality.

To prevent the specific information from being *mixed* across modalities, we impose a Total Correlation (TC) constraint on $\{U_m\}$:

$$TC(U_1, \dots, U_M) = D_{\text{KL}} \left(p(U_1, \dots, U_M) \left\| \prod_{m=1}^M p(U_m) \right\| \right), \quad (1)$$

where $p(\cdot)$ denotes the corresponding (joint or marginal) distribution, and $D_{\text{KL}}(\cdot \| \cdot)$ is the KL divergence. This measure characterizes the extent to which the joint distribution deviates from the product of marginals, and $TC = 0$ if and only if all U_m are mutually independent. The first part of the overall objective is written as:

$$L_1 = \sum_{m=1}^M \text{MSE}(h_m, \hat{h}_m) + \alpha TC(U_1, \dots, U_M) - \beta \sum_{m=1}^M I(C; h_m), \quad (2)$$

where $\text{MSE}(\cdot, \cdot)$ is the mean-squared-error reconstruction term, $I(\cdot; \cdot)$ denotes mutual information, and α, β are hyperparameter weights. The term $-\beta \sum_m I(C; h_m)$ can be understood as a *common information bottleneck*, encouraging C to maintain strong dependence with each modality representation h_m , so that C captures the signals consistent across all modalities. The second part takes $Z = [U_1, \dots, U_M, C]$

(concatenating all specific factors and the shared factor into a joint representation) and feeds it into a regression head to predict age, e.g., outputting the mean $\mu(Z)$, and minimizes $L_2 = \mathbb{E}(Y - \mu(Z))^2$, where Y is the ground-truth age label, and $\mathbb{E}$ denotes the expectation over the data distribution or a mini-batch. Finally, $L_{\text{total}} = L_1 + L_2$.

The key is that in high-dimensional spaces, $I(C; h_m)$ and $TC(U_1, \dots, U_M)$ are difficult to estimate stably. The paper uses two quantities induced by the Cauchy–Schwarz (CS) divergence to address this in a unified manner: it uses CS-QMI to estimate the mutual-information terms, and CS-TC to estimate the total-correlation term, thereby avoiding variational lower-bound approximations and not requiring any parametric assumptions on the distributions.

Concretely, the method constructs kernel Gram matrices on a mini-batch. Let the batch size be N , and denote the modality-specific representation of the k -th modality for the i -th sample by u_k^i . For each k , construct

$$Q_k(i, j) = G_\sigma(u_k^i - u_k^j), \quad (3)$$

where $Q_k \in \mathbb{R}^{N \times N}$ is the Gram matrix, $i, j \in \{1, \dots, N\}$ are sample indices, and $G_\sigma(\cdot)$ is a Gaussian kernel with bandwidth σ . CS-TC provides a log-ratio estimate formed from $\prod_k Q_k$, and it can be computed via matrix operations with complexity $O(MN^2)$, making the TC constraint differentiable and suitable for end-to-end training.

For the bivariate pair (C, h) , we similarly construct Gram matrices $K, P \in \mathbb{R}^{N \times N}$ (corresponding to kernel similarities of C and h , respectively). CS-QMI also yields a three-term log-ratio form, and can be expressed as a combination of traces and vector quadratic forms, such as $\text{tr}(KP)$, $\mathbf{1}^\top K \mathbf{1}$, $\mathbf{1}^\top K P \mathbf{1}$, etc., where $\text{tr}(\cdot)$ is the trace operator and $\mathbf{1} \in \mathbb{R}^N$ is the all-ones vector. This enables mutual-information-related terms to be stably estimated within a batch and participate in backpropagation, thereby supporting the joint optimization of *common information enhancement and modality-specific information decoupling*.

Selected Experimental Results

Experiments show that explicitly modeling common information can stably improve brain-age prediction: on REST-meta-MDD, MAE = 6.32 and Corr = 0.73; on ADNI, MAE = 5.33 and Corr = 0.50. The information decomposition indicates that, in MDD, the common information is 0.605, and the unique information of sMRI (0.578) is substantially higher than that of fMRI (0.023); in ADNI, the common information is 0.445, and the unique information of sMRI and fMRI are 0.340 and 0.208, respectively. The decomposition converges dynamically, and the modality contributions are consistent with neuroimaging evidence.